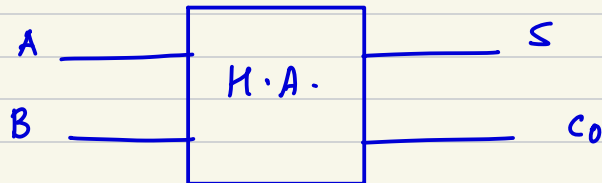




TRUTH TABLES, K-MAPS & SIMPLIFIED BOOLEAN EXPRESSION

1. Half Adder



Truth Table

A	B	S	C ₀
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

K-Map for Sum

		B	
		0	1
A	0		1
	1	1	

$$S = \overline{A}B + A\overline{B} = A \oplus B$$

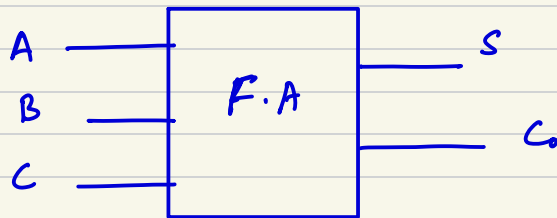
→
Simplified Boolean exp.

K-Map for C_0

A \ B	0	1
0	0	1
1	0	1

$$C_0 = AB$$

2. Full Adder



Truth Table

A	B	C	S	C_0
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

K-Map for Sum (S)

A \ BC	00	01	11	10
0		1		1
1	1		1	

0 1 3 2
4 5 7 6

$$S = A\bar{B}\bar{C} + \bar{A}\bar{B}C + ABC + \bar{A}B\bar{C}$$

$$S = A \oplus B \oplus C$$

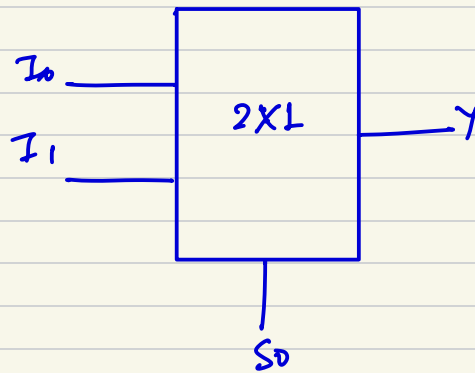
K-Map for C_0

A \ BC	00	01	11	10
0			1	
1		1	1	1

0 1 3 2
4 5 7 6

$$C_0 = BC + AC + \underline{\underline{AB}}$$

3. 2:1 MUX



Truth Table

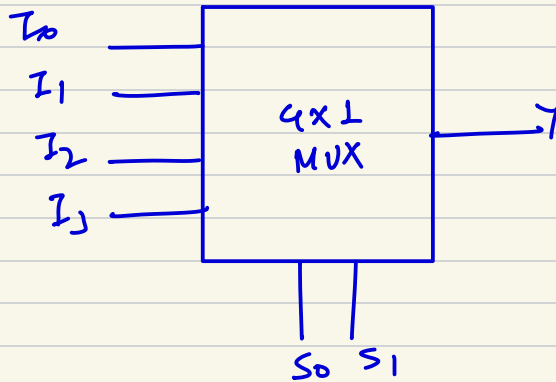
S_0	I_0	I_1	Y
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

$S_0 \backslash I_0 I_1$	00	01	11	10
0			1	1
1		1	1	

Karnaugh map for the 2:1 MUX output Y. The map shows the output for combinations of I_0 and I_1 for each value of S_0 . Red boxes highlight the groups of 1s: a group of two 1s for $S_0=0$ (minterms 3 and 2), a group of two 1s for $S_0=1$ (minterms 1 and 3), and a group of four 1s (minterms 3, 2, 1, 3).

$$Y = \overline{S_0} I_0 + S_0 I_1$$

4. 4x1 MUX



S_1	S_0	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

Boolean expression for Y

$$Y = \overline{S_1} \overline{S_0} I_0 + \overline{S_1} S_0 I_1 + S_1 \overline{S_0} I_2 + S_1 S_0 I_3$$